

Deepak Sathyan

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📍 Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

Education

University of Maryland

Ph.D. in Physics

Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass

Advisor: Dr. Kaustubh Agashe

College Park, MD

Aug 2018 – July 2024

Boston University

Bachelor of Arts in Physics, Summa Cum Laude with Honors

GPA: 3.9

Minor: Mathematics

Thesis: Muon ($g-2$): Searching for the Muon Electric Dipole Moment

Advisor: Dr. Robert Carey

Boston, MA

Aug 2014 – May 2018

Experience

Postdoctoral Research Associate

Mitchell Institute for Fundamental Physics and Astronomy

Texas A&M University

College Station, TX

Sept 2024 – Aug 2027

3 years

Selected Publications

Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm

2021

Muon $g-2$

arxiv.org/abs/2104.03281 (Phys. Rev. Lett.)

New laboratory constraints on neutrinophilic mediators

2025

P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang

arxiv.org/abs/2407.12738 (Phys. Lett. B)

LHC Signals for KK Graviton from an Extended Warped Extra Dimension

2020

K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan

arxiv.org/abs/2008.06480 (JHEP)

Publications

Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering

May 2026

B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan

arxiv.org/abs/2605.08010

A Baryon and Lepton Number Violation Model Testable at the LHC

Aug 2025

A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan

arxiv.org/abs/2508.21064

New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis

July 2025

P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang

arxiv.org/abs/2507.01000

New laboratory constraints on neutrinophilic mediators P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang arxiv.org/abs/2407.12738 (Phys. Lett. B)	2025
‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m_W K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan arxiv.org/abs/2404.17574 (JHEP)	2025
A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan arxiv.org/abs/2310.13687 (Phys. Lett. B)	2024
Unifying Searches for New Physics with Precision Measurements of the W Boson Mass D. Sathyan drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)	2024
Energy-peak based method to measure top quark mass via B-hadron decay lengths K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan arxiv.org/abs/2212.03929 (JHEP)	2023
TF07 Snowmass Report: Theory of Collider Phenomena Collaboration arxiv.org/abs/2210.02591	Oct 2022
Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021 Collaboration arxiv.org/abs/2209.13128	Sept 2022
Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021 Collaboration arxiv.org/abs/2209.11267	Sept 2022
Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan arxiv.org/abs/2204.02928	Apr 2022
Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan arxiv.org/abs/2203.13305	Mar 2022
The straw tracking detector for the Fermilab Muon g-2 Experiment Collaboration arxiv.org/abs/2111.02076 (JINST)	2022
Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm Muon g-2 arxiv.org/abs/2104.03281 (Phys. Rev. Lett.)	2021

Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab Muon g-2 arxiv.org/abs/2104.03240 (Phys. Rev. Accel. Beams)	2021
LHC Signals for KK Graviton from an Extended Warped Extra Dimension K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan arxiv.org/abs/2008.06480 (JHEP)	2020

Projects

BSM Searches via W Boson Mass Developed a framework unifying beyond-the-Standard-Model searches with precision measurements of the W boson mass at the LHC.	Jan 2020 – July 2024
Muon g-2 Experiment Contributed to the straw tracking detector and beam dynamics corrections for the Fermilab Muon g-2 experiment, which measured the muon anomalous magnetic moment to 0.46 ppm.	Aug 2018 – Jan 2022

Awards

Breakthrough Prize in Fundamental Physics Muon g-2 Collaboration	Apr 2026
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Skills

High Energy Physics

Computing